

Cryptocurrency Market Intelligence: End-to-End Data Analytics & Data Science Project Blueprint

1. Dataset Understanding (Verified by Direct Inspection)

I loaded and profiled all 23 files rather than trusting column names alone. Findings below are **confirmed facts**, not assumptions.

Structure: 23 separate CSVs, one per cryptocurrency, each with identical schema: **SNo, Name, Symbol, Date, High, Low, Open, Close, Volume, Marketcap**. This is daily OHLCV + market-cap data — a **multi-asset financial time-series panel**, not a cross-sectional or transactional dataset.

Fact	Value
Coins covered	23 (Bitcoin, Ethereum, XRP, Litecoin, Dogecoin, Monero, Stellar, Tether, NEM, IOTA, EOS, Binance Coin, TRON, Chainlink, Cardano, USD Coin, Crypto.com Coin, Wrapped Bitcoin, Cosmos, Solana, Polkadot, Uniswap, Aave)
Total rows across files	33,088
Global date range	2013-04-29 (Bitcoin, Litecoin) → 2021-07-06 (all coins, same end date)
Null/missing cells	0 in every file
Duplicate rows	0 in every file
Duplicate dates within a coin	0

This is an unbalanced panel, not a clean rectangular one. Every coin's series ends on 2021-07-06, but start dates are staggered by listing date (Bitcoin/Litecoin from 2013 down to Aave from Oct 2020). Only **275 days (2020-10-05 → 2021-07-06)** have all 23 coins present simultaneously. This single fact drives most of the cleaning and modeling design below — it is a

form of **survivorship/listing bias**: coins that exist in this dataset today all "survived," and cross-coin comparisons over the full history are only valid for the subset of coins alive at each point in time.

Real data-quality issues found (not assumed):

- **Zero trading volume**: Bitcoin and Litecoin each show exactly 242 zero-volume days, all clustered in their earliest history (2013) — this reflects incomplete exchange-volume reporting in early crypto data feeds, not literal zero trading.
- **Zero market cap**: Wrapped Bitcoin (196 days), Solana (52), Cosmos (46), TRON (14), Polkadot (12), USD Coin (8), Crypto.com Coin (2), EOS (1) — early-history rows where circulating-supply data wasn't yet tracked by the source. Market cap = Close × circulating supply, so a zero here means missing supply data, not a real \$0 valuation.
- **Small date gaps**: Monero is missing 1 calendar day; Tether is missing 5. Immaterial but must be handled explicitly (not silently forward-filled) before any time-series work.
- **No OHLC logic violations**: zero cases of High < Low, Close outside [Low, High], or negative prices/volumes across all 23 files — the core price data is internally consistent.
- **Extreme single-day moves (>50%)**: flagged in Dogecoin (11 days), Stellar (9), XRP (9), NEM (6), TRON (6), Binance Coin (5), and others. Some of these are real (e.g., the 2021 Dogecoin rally), some may be data artifacts — each needs individual verification, not blanket removal.
- **SNo is a per-file row counter, not a global key**. The real primary key is (Symbol, Date).

Structural facts that shape the whole project (confirmed by computation, not assumption):

- Annualized volatility of daily returns ranges from **8.8% (USD Coin)** and **9.7% (Tether)** up to **216.8% (Dogecoin)** — a ~20x spread. Stablecoins are a categorically different asset class in this data, purely from price behavior, with no label telling us so.
- Wrapped Bitcoin's return correlation with Bitcoin is **1.00** (it's a pegged, collateralized token — this is expected and a good sanity check).
- Over the shared 275-day window, Bitcoin's average correlation with the other 22 coins is **~0.5**, and its correlation with Tether/USDC is **~0 (slightly negative)** — quantitative confirmation that "crypto" is not one homogeneous asset class.
- Market-cap concentration on the last date is extreme: **Bitcoin + Ethereum = 72.9%** of the combined market cap of all 23 coins; the **top 5 coins = 85.5%**. A classic Pareto pattern.

What this dataset is NOT good for: it has no on-chain data, no transaction-level/wallet data, no news/sentiment/text, no macro variables, and no labeled outcome variable. Any "prediction" target must be engineered from price itself (e.g., future return direction), which means predictive claims must be handled with real statistical caution — financial daily-return prediction is a low-signal problem, and I flag this explicitly rather than overselling it.

2. What This Dataset Is Actually Useful For

Analytics type	Fit	Why
Descriptive	Strong	Price history, growth, volatility, drawdowns across 23 assets and ~8 years
Diagnostic	Strong	Why did volatility/correlation regimes shift (bull run, listing effects, stablecoin design)
EDA	Strong	Rich multivariate, temporal, and cross-sectional structure
Inferential/statistical	Moderate-Strong	Can rigorously test volatility/correlation differences between asset classes
Predictive (price forecasting)	Weak-Moderate	Daily price/return prediction is a near-random-walk problem; framed as a methodology exercise, not a trading system
Prescriptive	Moderate	Portfolio/diversification implications can be drawn, with appropriate caveats
ML — Clustering	Strong	Can the model rediscover "stablecoin vs. large-cap vs. altcoin" structure from price behavior alone, with no labels? Yes — validated above by hand. This is the standout ML opportunity.
ML — Classification/Regression	Weak	Next-day direction/return prediction; useful as a rigorous pipeline exercise (baselines, leakage prevention, walk-forward validation) but expected performance is close to chance and must be reported honestly
ML — Forecasting	Moderate	Volatility and price forecasting via classical time-series methods, evaluated with strict walk-forward validation
NLP / Recommendation / Ranking	Not applicable	No text or user-interaction data exists

Decision: this project is EDA + statistics + unsupervised ML (clustering) first, with a clearly-scoped, honestly caveated supervised/forecasting extension. Forcing a

high-confidence "predict tomorrow's price" model onto this data would misrepresent what daily crypto-price data can support.

3. Recommended Project

Project Title

"CryptoLens: Multi-Asset Market Structure, Volatility, and Risk Analytics for 23 Cryptocurrencies (2013–2021)"

Problem Statement

A retail investor, analyst, or fund allocator viewing "crypto" as an asset class needs to understand that it is *not* one thing—it spans near-zero-volatility stablecoins to 200%+ annualized-volatility speculative altcoins, with widely varying correlation to Bitcoin. This project builds a data-driven, evidence-based map of that structure: which coins behave alike, how risk and concentration evolved through the 2020–2021 bull run, and what a rigorously tested (not overhyped) predictive model can and cannot tell us about next-day price movement. It supports decisions like portfolio diversification, risk-tiering of assets, and setting realistic expectations for algorithmic crypto-trading claims.

Who uses it: retail/analyst investors doing due diligence, a data-science portfolio reviewer evaluating technical skill, or a fintech analytics team building a first-pass risk-tiering tool.

Business Questions

Basic

1. How have the price and market cap of each coin evolved over its listed history?
2. Which coins have the highest/lowest historical volatility?
3. What are the largest daily gainers/losers on record for each coin?

Intermediate 4. How correlated are coins with Bitcoin and with each other, and has that correlation been stable or has it shifted? 5. How did the Nov 2020–May 2021 bull run affect volatility and drawdowns differently across coins? 6. Do stablecoins (Tether, USD Coin) actually behave statistically differently from every other coin, or is that just visual intuition?

Advanced 7. Can an unsupervised model recover a sensible risk-tier segmentation (stablecoin/large-cap / high-beta altcoin) using only price-derived features, with no labels? 8. Is next-day return direction predictable above a naive baseline using lagged technical features, and how much of any apparent edge survives a leakage-safe, walk-forward validation? 9. How

concentrated is total market cap among the top coins over time (Pareto/HHI), and how has that concentration trended?

Executive/business 10. If you were building a "crypto risk tier" for a product (e.g., "conservative/moderate/aggressive" allocation buckets), which coins would land in which tier, and how confident should we be in that grouping? 11. What is the realistic, defensible claim this dataset supports about predicting crypto prices — and what claim would be overselling the data?

4. End-to-End Workflow (only the stages that earn their place)

Data Acquisition (already provided)

- Data Understanding & Profiling [done above]
- Data Quality Assessment
- Data Cleaning
- Feature Engineering
- Exploratory Data Analysis (univariate → multivariate → temporal → segmentation)
- Statistical Analysis (correlation, hypothesis tests)
- Advanced Analytics (volatility regimes, drawdown, Pareto/concentration, cohort-by-listing-date)
- Machine Learning: Clustering (core) + Direction Classification (caveated extension)
- Model Evaluation & Interpretation
- Business Insights
- Dashboard

Deployment/production ML serving is deliberately excluded from the core project — this is historical market data with no live feed, so "deployment" would be simulated rather than real; it's listed only as an optional advanced extension (Section 19).

5. Data Cleaning Plan

Issue

Decision

Reasoning

Zero-volume rows (early BTC/LTC)	Keep, flag with <code>is_zero_volume</code> indicator column. Do not impute or drop.	These are real historical rows; volume=0 likely means "not reported," not "no trading." Deleting shrinks legitimate early-history signal; imputing invents data. Flag lets EDA/ML exclude or weight them explicitly.
Zero-marketcap rows	Keep, flag with <code>is_zero_marketcap</code>. For marketcap-dependent analysis (Pareto, concentration), exclude flagged rows rather than treat as \$0.	A \$0 marketcap is not a real economic zero — including it would corrupt concentration metrics.
Small date gaps (Monero, Tether)	Leave as true gaps; do not forward-fill silently. For rolling-window features, use time-aware windows (<code>.rolling('7D')</code> on the actual date index) instead of row-count windows, so a gap doesn't quietly compress the window.	Silent forward-fill would fabricate a price/volume that never existed and bias volatility (a filled "no change" day artificially lowers measured volatility).
Unbalanced panel (staggered start dates)	Keep full history per-coin for single-asset analysis; separately build a "common window" subset (2020-10-05→2021-07-06, all 23 coins) for any cross-coin comparison, correlation, or clustering work.	Comparing volatility of a coin with 8 years of history to one with 9 months on unequal footing is misleading; the common window equalizes the comparison. This must be stated explicitly, not glossed over.
Extreme single-day moves (>50%)	Do not remove. Cross-check each against known events (e.g., Dogecoin's 2021 rally) where feasible; keep as legitimate extreme observations for volatility/anomaly analysis, since crypto genuinely has fat-tailed, non-normal returns.	Removing "outliers" in a market notorious for real 50%+ days would delete the exact behavior we're studying — this is a case where "outlier" ≠ "error."
<code>SNo</code> column	Drop for analysis; it's a per-file row index, not a meaningful ID. Use (<code>Symbol</code> , <code>Date</code>) as the true composite key.	Prevents accidental joins/keys on a meaningless counter.

Timestamp granularity (23:59:59 on every row)	Truncate to date-only (<code>normalize()</code>).	It's daily data with a constant end-of-day timestamp; keeping the time component adds noise with zero information.
Data types	Cast <code>Date</code> to <code>datetime</code>, numeric columns confirmed already float/int — no coercion issues found.	Verified directly; no silent type problems.
Constant/near-constant columns	None found (<code>Symbol/Name</code> are constant <i>within</i> a file, which is expected and fine — they become the join key across the combined dataset).	—

6. Feature Engineering Plan

Feature	Formula / Logic	Purpose	Expected value
Daily return	<code>Close.pct_change()</code>	Core unit for volatility/correlation work (price levels aren't comparable across coins of vastly different price scale)	High
Log return	<code>log(Close/Close.shift(1))</code>	Statistically better-behaved for volatility modeling than simple returns	High
Rolling volatility (7d, 30d)	<code>return.rolling('30D').std() * sqrt(365)</code>	Captures time-varying risk, not just one full-sample number	High
Drawdown	<code>Close / Close.cummax() - 1</code>	Measures worst-case loss from peak — directly answers a risk question raw price can't	High

Intraday range	<code>(High - Low) / Open</code>	Proxy for intraday volatility/liquidity, independent of close-to-close return	Medium
Volume z-score (rolling)	<code>(Volume - roll_mean) / roll_std</code>	Flags unusual trading activity days, feeds anomaly analysis	Medium
Market-cap share	<code>Marketcap_coin / sum(Marketcap_all coins that day)</code>	Enables the Pareto/concentration analysis (Q9)	High
Days since listing	<code>Date - first_date_for_coin</code>	Lifecycle/cohort feature — controls for "young coin" effects (e.g., new listings are often more volatile)	Medium
Lagged returns (1, 3, 7-day)	<code>return.shift(1..7)</code>	Candidate predictors for the classification extension	High (for ML section only)
RSI, moving-average crossover	Standard technical-indicator formulas on <code>Close</code>	Common, interpretable technical features for the classification baseline	Medium
<code>is_zero_volume</code> , <code>is_zero_market cap</code>	Boolean flags from Section 5	Data-quality flags carried into modeling so bad rows can be excluded/weighted rather than silently biasing results	High

Explicitly avoided: target encoding, extensive interaction terms, or dozens of technical indicators — with only price/volume/marketcap available, most "exotic" TA features would be redundant transforms of the same three underlying series and would mainly add noise and overfitting risk to the classification extension.

7. Exploratory Data Analysis Plan

Univariate — per-coin price/return/volume distributions; return skew/kurtosis (expect fat tails, confirmed by the >50%-move counts above) → answers "how risky does this single asset look in isolation?"

Bivariate — coin-vs-coin return correlation heatmap (common window); Bitcoin-as-benchmark scatter (each coin's daily return vs. BTC's) → answers "what moves together, and does anything move independently (hedging candidates)?"

Multivariate — PCA on the common-window return matrix (23 coins × 275 days) to see how many independent "factors" drive crypto markets, and whether stablecoins load on a separate factor from everything else → answers "is 'crypto' really 3–4 underlying risk factors wearing 23 different tickers?"

Temporal — full-history price/volatility line charts faceted by coin cohort (early: BTC/LTC/XRP/DOGE; mid: ETH/NEM/Tether-era; late: DeFi-era SOL/DOT/UNI/AAVE); rolling 30-day volatility overlay during the Nov 2020–May 2021 run-up vs. the May–Jul 2021 correction → answers "did the bull run affect all coins equally, or did risk profiles diverge?"

Segmentation — the clustering output from Section 10, visualized as a scatter of (volatility, BTC-correlation) colored by cluster → directly answers Q7 and Q10.

Anomaly — table of every >50% single-day move, cross-tabulated with the `is_zero_volume/is_zero_marketcap` flags to separate "real market anomaly" from "data-quality artifact" → protects every downstream number from silently including bad rows.

For each chart the same discipline applies: state the question it answers and the decision it could inform, not just "here's a plot."

8. Statistical Analysis Plan

Test	Hypothesis	Assumptions checked	Business meaning
Levene's / F-test on return variance	H0: stablecoins (USDC, Tether) have the same return variance as the median altcoin	Independence of daily returns (approximately reasonable at daily frequency); will report if violated	Formally confirms — rather than just visually asserts — that "stablecoin" is a statistically distinct risk category, backing the risk-tiering business question

Welch's t-test on mean daily return, bull run vs. correction period	H0: mean daily return is equal in the two regimes	Uses Welch's version specifically because variances differ across regimes (already known to be unequal)	Quantifies whether the "correction" was a statistically significant regime shift or noise
Pearson correlation + significance, each coin vs. BTC	H0: true correlation is 0	Requires roughly linear relationship; will inspect scatter for violations	Distinguishes coins that are statistically independent of BTC (diversification value) from those that aren't
Augmented Dickey-Fuller test on price levels vs. returns	H0: series has a unit root (non-stationary)	Standard for time-series	Confirms <i>why</i> returns, not raw price, must be used for correlation/volatility work — raw crypto prices are non-stationary, so a naive price-level correlation would be spurious

Correlation ≠ causation, stated explicitly: e.g., Wrapped Bitcoin's 1.00 correlation with Bitcoin is a *causal/mechanical* fact (WBTC is contractually pegged to BTC), while a moderate BTC–altcoin correlation is an *observed association* driven by shared market sentiment, exchange flows, and macro liquidity — not evidence that BTC's price *causes* altcoin moves.

9. Advanced Analytics

Technique	Applied how	Answers
Pareto / concentration analysis	Market-cap share by coin over time; confirmed BTC+ETH = 72.9% , top-5 = 85.5% of combined cap on the final date	How concentrated is "the crypto market," and is that concentration rising or falling through the sample?
Drawdown / underwater analysis	Peak-to-trough % loss per coin across full history	What's the realistic worst-case loss an investor in each coin has actually experienced?

Cohort-by-listing-date analysis	Group coins by the year they entered the dataset (2013 cohort vs. 2017 cohort vs. 2020 "DeFi era" cohort) and compare typical volatility/return profiles	Does a coin's "era" of listing predict its risk profile (e.g., are 2020–21 DeFi-era listings systematically more volatile than 2013-era coins)?
Rolling correlation (regime) analysis	90-day rolling correlation of each altcoin with BTC	Is "crypto moves together" a stable law or a time-varying phenomenon that intensifies in certain regimes (e.g., risk-off periods)?

Explicitly not used: market-basket analysis, funnel analysis, RFM, survival analysis, geographic/network analysis — none apply; this is single-asset price data with no transactions, customers, or geography.

10. Machine Learning Recommendation

Two ML components, deliberately scoped differently:

10a. Core model — Unsupervised Clustering (highest-confidence ML story)

- **Problem:** cluster the 23 coins using price-derived features only (annualized volatility, BTC correlation, average daily volume relative to marketcap, max drawdown, days-since-listing) computed over the common 275-day window.
- **Why this is the strong ML choice:** we already *know* the answer well enough to sanity-check the model (stablecoins should isolate; WBTC should sit near BTC), which makes this a rare case where an unsupervised result is independently verifiable rather than "trust the algorithm."
- **Method:** K-Means (with elbow/silhouette to pick k) and hierarchical clustering as a cross-check, standardizing features first (volatility and volume-ratio are on very different scales).
- **Baseline:** a naive 2-cluster split by volatility alone (>50% vs. <50% annualized) — the full model must beat this simple rule to justify its complexity.
- **Evaluation:** Silhouette score, Davies–Bouldin index, and — critically — **face-validity check:** does Tether/USDC land alone or together in their own cluster? This qualitative check matters as much as the numeric score here.

10b. Extension — Next-Day Direction Classification (framed with explicit honesty)

- **Target:** 1 if next day's Close > today's Close, else 0, per coin, using the common window (or full per-coin history for coins with enough data).
 - **Baseline:** predict the majority class, and separately, "predict yesterday's direction repeats" — both must be beaten for any model to claim value.
 - **Features:** lagged returns, rolling volatility, RSI, moving-average crossover signal — all computed using only past data relative to the prediction date (see Section 11).
 - **Models:** Logistic Regression and Random Forest only. **XGBoost/LightGBM/deep learning are deliberately NOT used here** — with ~10 features and a target that's close to a coin-flip in efficient markets, heavier models mainly risk overfitting noise, not capturing real signal; a data scientist should be able to justify *not* reaching for the fanciest tool.
 - **Expected outcome, stated upfront:** accuracy is expected to land close to 50–55%. **This is reported as the honest finding**, not hidden — the real deliverable here is a leakage-safe, walk-forward-validated pipeline, and the finding "raw daily OHLCV alone gives little edge over a coin flip" is itself a legitimate, defensible business insight (Q11).
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11. Leakage Prevention

- **Temporal leakage:** every feature for predicting day t uses only data through day $t-1$; enforced with `.shift(1)` before any rolling/lag feature is computed, and validated by asserting no feature's underlying window extends past the prediction date.
 - **Train/test split:** chronological, walk-forward (expanding window) — never a random shuffle split, which would let the model "see the future" via adjacent days in the training set.
 - **Cross-coin contamination:** when pooling coins for the classification model, splits are done by date globally, not by coin, and coin identity is never used as a leak-prone proxy for the target.
 - **Clustering:** since clustering has no "future" to leak from, the main risk is using the *full-history* end-state marketcap ranking as a feature — avoided by computing all clustering features only within the common evaluation window.
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12. Model Interpretation

- **Clustering:** interpret clusters via their centroid feature values (e.g., "Cluster 0: mean volatility 9%, mean BTC-correlation -0.05 → stablecoins") rather than treating cluster labels as black boxes; visualize with a 2D PCA or direct (volatility, correlation) scatter.
- **Classification:** Logistic Regression coefficients read directly; Random Forest via permutation importance and a **calibration check** (is a 55%-confidence prediction

actually right 55% of the time?) — critical in a low-signal problem where raw accuracy can be misleading.

- **Error analysis:** confusion matrix broken out by regime (bull run vs. correction) to check whether the model's few correct calls cluster in one market condition — a common failure mode where "skill" is actually just a lucky sub-period.

13. Model Comparison

Model	Strengths	Weaknesses	Key metric	Interpretability	Cost	Decision
K-Means (clustering)	Fast, simple, matches known ground truth well	Assumes spherical clusters; sensitive to k choice	Silhouette + face validity	High	Low	Selected — primary ML deliverable
Hierarchical clustering	No need to pre-pick k; dendrogram is itself an insight	Slower, less scalable (irrelevant at n=23)	Cophenetic correlation	High	Low	Used as a cross-check, not primary
Logistic Regression (direction)	Fully interpretable, fast, honest baseline-beater test	Assumes linear decision boundary	Accuracy, log-loss, calibration	High	Low	Selected — reports the realistic result
Random Forest (direction)	Captures nonlinearity/interactions	Larger overfitting risk on ~10 noisy features	Accuracy, log-loss	Medium	Medium	Selected as the "give it a fair chance" second model

XGBoost/deep learning	Could squeeze out marginal accuracy	High overfitting risk on this feature set; poor interpretability; hard to justify given expected near-chance signal	—	Low	High	Not selected — complexity not justified by the data
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14. Business Insights Framework

Every finding follows: **Finding** → **Evidence** → **Why it matters** → **Action**. Examples already grounded in the real numbers above:

- Finding:** Stablecoins are a categorically distinct risk class. **Evidence:** USDC/Tether annualized volatility 8.8–9.7% vs. 80–217% for every other coin; near-zero correlation with BTC. **Why it matters:** treating "crypto" as one risk bucket in a portfolio tool would be a material modeling error. **Action:** any risk-tiering or allocation tool built on this kind of data must hard-code stablecoins as a separate tier, not let a generic model discover it fresh each time (though Section 10a shows it *can*).
 - Finding:** Crypto market cap is highly concentrated. **Evidence:** BTC+ETH = 72.9% of total combined cap; top 5 = 85.5%. **Why it matters:** "diversifying across altcoins" provides much less true diversification of *capital exposure* than the coin count suggests. **Action:** portfolio-construction tools should weight by realized correlation/marketcap share, not just count of assets held.
 - Finding:** Daily-return direction is close to unpredictable from OHLCV alone. **Evidence:** baseline-beating classification accuracy expected only marginally above 50%. **Why it matters:** protects against overselling a trading-signal claim that the data doesn't support. **Action:** recommend against deploying this feature set as a live trading signal; recommend it as a template for testing *richer* data (order book, on-chain, sentiment) where a real edge is more plausible.
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15. Dashboard Recommendation

Yes, a dashboard adds value — this is inherently a "compare 23 time series and their relationships" problem, which benefits from interactivity (filter by coin, date range, and cluster).

- **Executive KPI layer:** total combined market cap (latest), BTC dominance %, average 30-day volatility across all coins, count of coins currently in "correction" (>20% drawdown from peak)
 - **Trend panel:** multi-select line chart of normalized price (indexed to 100 at each coin's listing) so coins of wildly different price scales are visually comparable
 - **Segment panel:** the cluster scatter (volatility vs. BTC-correlation), clickable to filter the trend panel
 - **Correlation panel:** heatmap with a date-range slider to show regime shifts
 - **Drill-down:** click a coin → single-asset detail view (price, volume, drawdown, rolling volatility)
 - **Tool recommendation:** **Streamlit or Plotly Dash**, not Power BI/Tableau — the clustering and rolling-window calculations are Python-native and benefit from being computed in the same codebase as the dashboard rather than re-implemented in a BI tool's formula language; Streamlit specifically for fastest build time on a portfolio timeline.
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16. SQL Layer

Worth including — realistic if the cleaned CSVs are loaded into a local SQLite/Postgres table `prices(symbol, date, open, high, low, close, volume, marketcap)`.

Representative tasks (not toy examples):

- Window function: 30-day rolling volatility per coin using `STDDEV(...) OVER (PARTITION BY symbol ORDER BY date ROWS BETWEEN 29 PRECEDING AND CURRENT ROW)`
 - CTE + ranking: for each date, rank coins by marketcap and extract the top-5 threshold to compute the Pareto share over time
 - Cohort query: `GROUP BY DATE_TRUNC('year', first_listed_date)` joined back to average volatility, to reproduce the cohort-by-listing-date analysis in SQL
 - Anomaly query: `WHERE ABS(close/LAG(close) OVER (PARTITION BY symbol ORDER BY date) - 1) > 0.5` to reproduce the extreme-move detection directly in SQL, as a cross-check against the pandas version
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17. Project Architecture

```
cryptolens/
|
|— data/
|  |— raw/          # original 23 CSVs, untouched
```

```

├── processed/           # cleaned + flagged, long-format combined table
├── notebooks/
│   ├── 01_data_understanding.ipynb
│   ├── 02_data_cleaning.ipynb
│   ├── 03_feature_engineering.ipynb
│   ├── 04_eda.ipynb
│   ├── 05_statistical_analysis.ipynb
│   ├── 06_advanced_analytics.ipynb
│   ├── 07_clustering.ipynb
│   └── 08_direction_classification.ipynb
├── src/
│   ├── data_processing.py    # load, clean, flag, combine 23 files into one long table
│   ├── feature_engineering.py
│   ├── analysis.py          # stats tests, drawdown, Pareto
│   └── modeling.py           # clustering + classification pipelines, walk-forward split
├── dashboard/
│   └── app.py                # Streamlit app
├── sql/
│   └── queries.sql
├── reports/
│   └── findings_summary.md
├── requirements.txt
├── README.md
└── .gitignore

```

18. Technology Stack

Layer	Tool	Why (not defaults)
Data	Pandas, NumPy	Standard, sufficient — no big-data tooling needed at 33K rows
Visualization	Plotly (primary), Matplotlib/Seaborn (static report figures)	Plotly needed for the interactive multi-coin comparisons; static libs fine for report-only charts

Statistics	SciPy, Statsmodels	Levene's, Welch's t-test, ADF test
ML	scikit-learn only	K-Means, hierarchical clustering, LogisticRegression, RandomForest — no XGBoost/LightGBM given the reasoning in Section 13
Database	SQLite	Lightweight, no server needed for a static historical dataset; Postgres is overkill here
Dashboard	Streamlit	Fastest path from Python analysis code to an interactive app
Deployment (optional)	Docker + Streamlit Community Cloud	Simple, free hosting path for a portfolio piece

19. Advanced Extensions (ranked)

Extension	Difficulty	Portfolio value	Practical value	Rank
Volatility forecasting (GARCH)	Medium	High	Medium	1
Deploy dashboard publicly (Streamlit Cloud)	Low	High	Medium	2
Automated data-refresh pipeline (pull fresh prices via API)	Medium	Medium	High	3
Portfolio-optimization module (mean-variance using the correlation matrix)	Medium	High	Medium	4
Price forecasting via Prophet/ARIMA per coin	Medium	Medium	Low (given weak predictability finding)	5
Model monitoring/retraining schedule	Medium	Low	Low (no live signal to monitor)	6

20. Beginner → Advanced Roadmap

Level 1 — Beginner: load and combine all 23 files; data cleaning with the flags from Section 5; univariate/bivariate EDA; basic SQL (Section 16, first two queries).

Level 2 — Intermediate: feature engineering (Section 6); full statistical testing (Section 8); Pareto/drawdown/cohort analytics (Section 9); Streamlit dashboard v1.

Level 3 — Advanced: clustering with interpretation (Section 10a); direction-classification pipeline with leakage-safe walk-forward validation and honest reporting (Section 10b); one advanced extension from Section 19.

21. Portfolio & GitHub Strategy

README should include: overview, business problem, dataset description (with the real facts from Section 1 — staggered listings, common window caveat, data-quality flags), methodology, key findings (the three from Section 14), dashboard screenshot, architecture diagram, install/usage instructions, limitations (Section 23), future improvements.

Resume description (concise, not exaggerated):

Built an end-to-end analytics pipeline on 8 years of daily price data across 23 cryptocurrencies (33K+ rows); identified and quantified a structural volatility gap between stablecoins and speculative assets (8–10% vs. 80–217% annualized volatility) via hypothesis testing; used K-Means clustering to recover this risk segmentation unsupervised; built a leakage-safe, walk-forward-validated classification pipeline for next-day price direction and reported its realistic (near-baseline) performance rather than overstating predictive power; delivered findings via an interactive Streamlit dashboard.

22. Interview Preparation

Data analysis: "Why did you flag zero-volume rows instead of dropping them?" → Dropping would remove real early-history rows just because volume wasn't reported; flagging preserves the data while letting downstream analysis exclude them where volume matters.

SQL: "Write a query to find each coin's max drawdown using window functions." → Use `MAX(close) OVER (PARTITION BY symbol ORDER BY date ROWS UNBOUNDED PRECEDING)` as running peak, then `close/peak - 1`, then `MIN(...)` per symbol.

Statistics: "How do you know the volatility difference is significant and not noise?" → Levene's test on variance equality across groups, reported with the actual test statistic and p-value, not just a visual gap.

ML: "Why not use XGBoost for the direction model?" → With ~10 features and a target close to random given market efficiency, a high-capacity model mainly overfits noise; a simpler model that's easier to validate and interpret is the more defensible choice, and the project's finding (near-baseline accuracy) doesn't change with a fancier algorithm.

Business: "What decision would you actually support with this?" → Risk-tiering assets for a portfolio tool and setting realistic expectations about what OHLCV-only signals can and can't predict — explicitly not "which coin to buy."

Architecture: "Why Streamlit over Power BI?" → The clustering/statistical outputs are computed in Python; Streamlit keeps analysis and presentation in one codebase instead of re-deriving logic in a BI tool.

Critical thinking: "What's the biggest limitation of your whole approach?" → The common-window cross-coin comparisons only cover ~9 months (Oct 2020–Jul 2021) — a single, unusually bullish market regime — so any cluster or correlation structure found may not generalize to a bear market or to a longer history.

23. Project Limitations (stated plainly)

- **Regime narrowness:** the fully-overlapping 275-day window used for all cross-coin comparisons is entirely within one strong bull market — findings about correlation/clustering structure may not hold in other regimes.
 - **Listing/survivorship bias:** the dataset only contains coins that were still trading as of mid-2021; delisted or failed coins are invisible, which likely makes the "risk" of the altcoin category look milder than the true historical base rate.
 - **No fundamental/on-chain/sentiment data:** limits both the ceiling on predictive modeling and the depth of "why" behind diagnostic findings.
 - **Zero-volume/zero-marketcap gaps concentrated in early history:** any full-history metric (e.g., "lifetime volatility") for Bitcoin/Litecoin is computed over a period that includes 242 zero-volume days each — worth a sensitivity check excluding those days.
 - **No causal claims:** correlation and clustering results describe association, not causation, and are explicitly not sold as such anywhere in this project.
 - **Direction-classification results are expected to be weak by design of the problem, not a modeling failure** — this is disclosed rather than framed as underperformance to fix.
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24. Prioritized Execution Roadmap

Priority	Task	Impact	Difficulty
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Must have	Combine + clean all 23 files with quality flags	High	Low
Must have	Core EDA (univariate/bivariate/temporal)	High	Low
Must have	Statistical tests (stablecoin variance, BTC correlations)	High	Medium
Must have	Pareto/concentration + drawdown analytics	High	Medium
Should have	Clustering with interpretation	High	Medium
Should have	Streamlit dashboard v1	Medium	Medium
Should have	SQL layer	Medium	Low
Advanced	Leakage-safe direction classification	Medium	High
Advanced	Cohort-by-listing-date analysis	Medium	Medium
Optional	GARCH volatility forecasting	Low-Medium	High
Optional	Public deployment + auto-refresh pipeline	Low (portfolio polish)	Medium

25. Summary of Real, Verified Findings Used Throughout This Blueprint

- 23 coins, 33,088 rows, 2013-04-29 → 2021-07-06, zero nulls/duplicates, but real zero-volume/zero-marketcap data-quality gaps in early history.
- Only 275 days have all 23 coins simultaneously listed — the hard constraint behind the "common window" design choice used everywhere in Sections 5–10.
- Annualized volatility spans 8.8% (USD Coin) to 216.8% (Dogecoin).
- Wrapped Bitcoin–Bitcoin correlation = 1.00; Bitcoin's average correlation with other coins ≈ 0.5 ; with stablecoins ≈ 0 .
- BTC + ETH = 72.9% of combined market cap on the final date; top 5 coins = 85.5%.

Every design decision above traces back to one of these numbers rather than a generic template — that traceability is itself the strongest thing to point to in an interview.